Dwell-Adaptive RF Signal Classification from Duration-Complete Synthetic Data

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Abstract

Burst-structured radio protocols—TDMA cellular, frequency-hopped and sparse-advertising Bluetooth—are routinely misclassified by deep I/Q classifiers because the identity of these signals lives at millisecond timescales that fixed-sample-count capture windows rarely contain. We present DACS (Dwell-Adaptive Classification from duration-complete Synthesis), a method with three components: a synthesis principle, in which training captures are windows of index-pure signal processes composed from standards-derived packet waveforms on unbounded slot- and event-clocked timelines, so captures span durations commensurate with each protocol's defining timescales, silence enters training as class-conditional evidence rather than being rejected, and every capture is emitted as a clean/impaired pair; a multi-objective network, in which a spectro-temporal encoder feeds a prototype head, a masked denoising decoder supervised by the clean pair, a worst-case-over-dwell loss, and a learned confidence signal for an answer-early/escalate dwell policy; and an evaluation-design analysis showing that minimum-over-cells metrics are order statistics whose readout sits far below true per-cell recall at small cell sizes, with a cell-size prescription that restores fidelity. In controlled experiments holding architecture, optimizer, and training budget fixed, duration completeness alone lifts 7-class balanced accuracy from 0.503 (shortest fixed-count window tier) to 0.954 at 1 ms dwell and to 1.000 at 10 ms, replicated on two seeds; neither architecture substitution, longer training, nor worst-case reweighting on duration-starved data approaches this effect. Training cost is linear in dwell.

Index terms—RF signal classification, synthetic training data, frequency hopping, selective classification, anytime inference.

1 Introduction

Deep networks classify radio emissions directly from sampled I/Q [1, 2], and for continuously transmitting waveforms—tones, analog modulations, OFDM carriers—short captures suffice: their signatures are spectral and quasi-stationary. Burst-structured protocols break this regime. A GSM carrier transmits one 577 μs burst per 4.615 ms TDMA frame [14]; Bluetooth Classic occupies one of 79 channels for one 625 μs slot before hopping; BLE advertising transmits three sub-160 μs packets every few tens of milliseconds [13]. Within any window shorter than these periods, all three collapse to "a narrowband blip or nothing" (Fig. 1)—and a classifier trained on such windows inherits the ambiguity no matter its architecture.

The prevailing practice of fixing capture length in *samples* while sweeping sample rate makes this failure systematic: at high rates a fixed-count window spans microseconds of physical time. We identify and quantify three consequences (Sec. 3): captures of bursty classes are frequently near-empty; class-defining temporal structure exceeds the observation span even when energy is present; and the worst-cell evaluation metrics common in deployment contracts are statistically unable to display high performance. These observations motivate a method built around *duration*:

- **Duration-complete paired synthesis** (Sec. 4): index-pure generative processes replay standards-derived packet waveforms on unbounded hop/event timelines, yielding captures of arbitrary physical duration with byte-exact block composition, silence included at its true class-conditional rate, and every capture paired with its pre-impairment reference.
- **A dwell-adaptive multi-objective network** (Sec. 5): prototype classification [3] over episodes, masked denoising reconstruction against the clean pair [5, 4], a worst-case-over-dwell loss, and a learned confidence signal designed to convert fixed-dwell classification into an escalation policy [8, 10].
- **Worst-cell evaluation design** (Sec. 3): a Monte Carlo analysis of minimum-over-cells metrics as order statistics, with a prescription for cell sizes that make ≥0.9 scores displayable.

Controlled experiments (Sec. 6) isolate each factor. With architecture, recipe, and budget held fixed, replacing fixed-count training data with duration-complete data moves 7-class balanced accuracy from 0.503 (shortest window tier) to 0.954 (1 ms dwell), and to 1.000 at 10 ms, on both tested seeds. Conversely, on the fixed-count corpus, architecture substitution fails to beat a feature-engineered baseline, tripling the training budget *reduces* worst-window balanced accuracy, and worst-case loss reweighting recovers at most +0.055: the data, not the model, is the binding constraint.

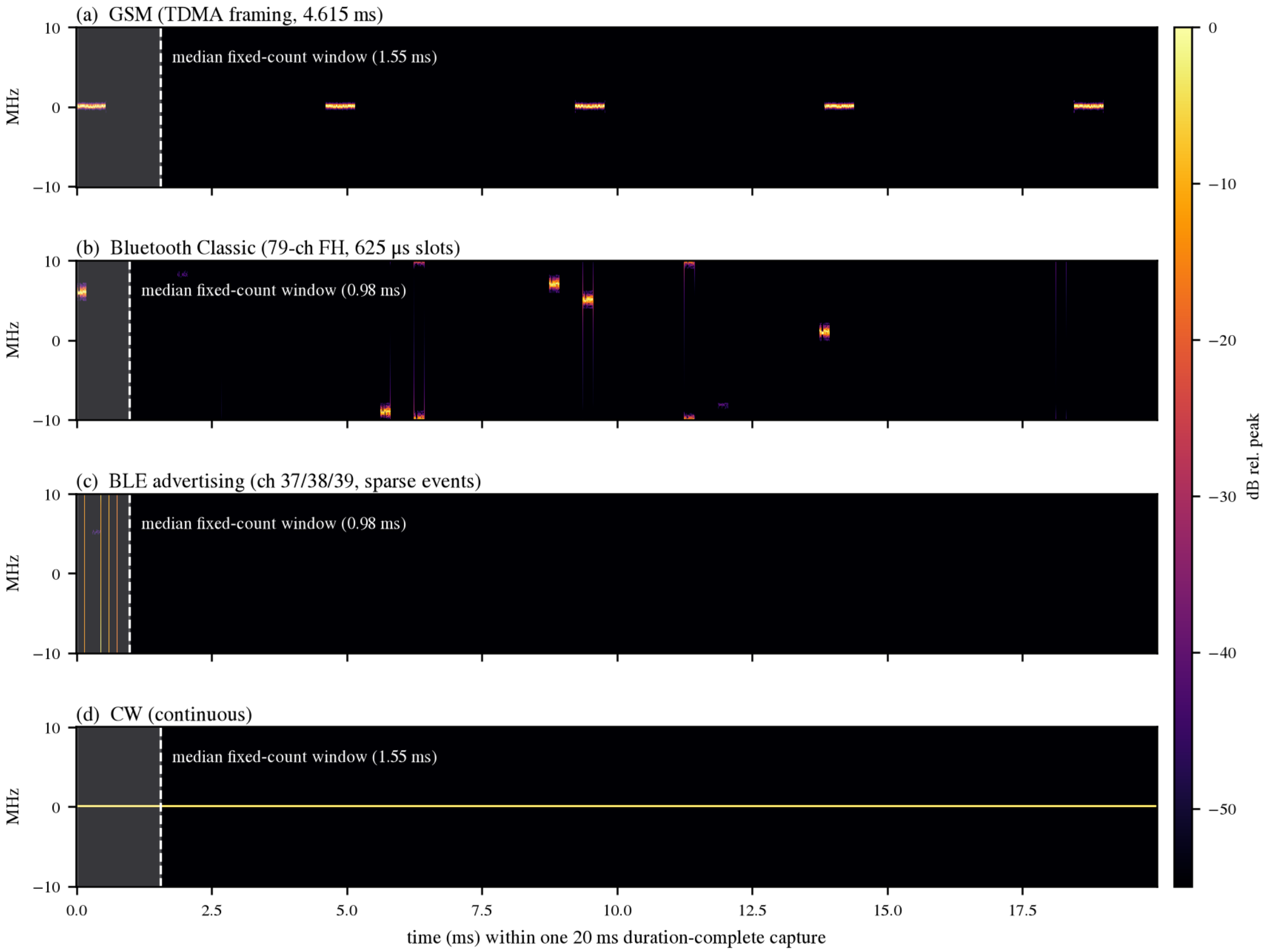

Figure 1: Class identity lives at timescales a fixed-sample-count window rarely contains. Spectrograms of single 20 ms duration-complete captures (clean synthesis, 20 Msps): (a) GSM, (b) Bluetooth Classic, (c) BLE advertising, (d) CW. Shading marks the median window obtained by the common practice of fixing capture length in samples while sweeping sample rate (here 16,384 samples over 1–200 Msps). Within that window, GSM shows at most one burst, Bluetooth Classic at most one hop, and BLE one event or pure silence. The 20 ms capture contains the discriminants: TDMA framing [14], slot-clocked hopping and sparse advertising events [13], continuity for CW.

2 Related Work

Convolutional and recurrent networks on raw I/Q established deep automatic modulation classification [1], and over-the-air studies exposed the sim-to-real and impairment sensitivity of such models [2]. Our contribution is orthogonal to architecture: we show that for burst-structured protocols the *temporal extent* of training captures is the first-order factor, and we supply the synthesis machinery that makes duration a free parameter. Prototype networks [3] provide the episodic head we retain for open-set enrollment; masked autoencoding [4] and denoising [5] motivate the reconstruction objective, which our clean/impaired pairs supervise exactly rather than by proxy. The dwell policy instantiates selective prediction [8, 7] and early-exit inference [10, 11] along a new axis: physical observation time rather than network depth, with calibration [9] governing the escalation thresholds.

3 Why Fixed-Count Windows Fail

We quantify the failure on the *fixed-count control corpus*: captures of 16,384 samples at rates swept over 1–200 Msps (median TDMA-class window 1.55 ms), constructed to instantiate the prevailing convention with the same classes and impairment envelope as our method's corpus. We call such data *duration-starved*. Throughout, *worst-window balanced accuracy* denotes the minimum over the corpus's three window tiers (4,096/8,192/16,384 samples) of 7-class balanced accuracy.

Bursty captures are near-empty. Against the stationary expectation that the first quarter of a capture holds 25% of its energy, 37% of GSM captures hold <10% of capture energy in their first 4,096 samples (Fig. 2a aggregates all GSM configurations; the burst-mode configurations individually range 40–48%, while the always-on broadcast carrier is ≈0%; block crest factor 39×); frequency-hopped captures 23%; DSSS 24%; continuous classes ≈0%. Conventional pipelines *reject* empty windows as invalid, discarding exactly the statistic that separates low-duty-cycle classes: $P(\text{near-empty} \mid \text{class})$. A prior-optimal treatment of empty windows alone bounds balanced accuracy near 0.93.

Structure exceeds the span. The continuously transmitting GSM configuration (loaded broadcast carrier)—never empty—still yields only 0.54 recall on the shortest window tier under the feature-engineered baseline of Sec. 6 (mean over its 24 tuned configurations): the GSM/Bluetooth discriminant is frame timing versus hop timing, both invisible inside a sub-millisecond span. Energy presence is not the limiting factor; temporal structure is.

The metric cannot display success. Acceptance criteria for deployed classifiers often take the form of minimum recall over (configuration×condition) cells. With J cells of n evaluation rows, the displayed value is $\mathbb{E}[\min_j \hat{p}_j]$ with $\hat{p}_j \sim \mathrm{Bin}(n,p)/n$; for a representative design with J=124 cells of n=16 (e.g., 31 configurations crossed with 4 conditions), true per-cell recall 0.95/0.97/0.99 reads as 0.77/0.82/0.89 (Fig. 2b). A system can be near-ceiling while the metric reports 0.8. The remedy is not a different statistic but adequate cells: at J=34, n=96, a true 0.97 displays as 0.93, and ≥0.9 scores become achievable in principle rather than statistically excluded. We adopt per-configuration cells of n=96 per dwell throughout, reporting balanced accuracy alongside the minimum cell.

4 Duration-Complete Paired Synthesis

Index-pure processes. Every generator is a deterministic function of the absolute sample index, $y[n] = G(n)$, with no hidden state across calls. Purity gives *stitch-exactness*: synthesizing $[0, N)$ in one call or any partition of consecutive calls yields identical bytes, so captures of arbitrary duration are assembled from bounded, individually validated content. Continuous and frame-cyclic waveforms satisfy purity natively; the contribution is extending it to aperiodic burst processes.

Burst-process composition. Let $b[\cdot]$ be a standards-derived baseband packet waveform of B samples (e.g., a digitally verified DH1 slot or BLE advertising PDU [13]). For a slot-clocked frequency hopper with slot length S, channel count K, and utilization ρ,

$$s = \lfloor n/S \rfloor, \qquad a(s) = \mathbb{1}[h_1(s) < \rho]\,,$$
$$y[n] = a(s)\, b[n - sS]\, \exp\big(j\, 2\pi \Delta f_{c(s)} n / f_s + j\phi(s)\big),$$

where h_1, h_2, ϕ are keyed integer hashes of the slot index, $c(s) = h_2(s) \bmod K$ selects the hop channel, and Δf_c is the channel offset from the capture center. Sparse advertising is composed analogously on an event grid of period P with per-event pseudo-random delay (the specification's *advDelay* model) and per-event multi-channel transmission. Both processes are pure in n, unbounded, and reproduce the *statistics* of channel occupancy and timing; the specification's hop-selection kernel and payload variation are explicitly out of scope and disclosed as such.

Pairs and impairments. Stage one synthesizes clean captures at each waveform's native rate (1.3–122.88 Msps here), block-stitched; stage two applies a rational polyphase resampler to a common rate, then a seeded receiver chain—AWGN at $U(-5, 45)$ dB SNR referenced to *active-sample* power (SNR of a burst is a property of the emission, not of inter-burst silence), LO-scaled carrier offset, Wiener phase noise, IQ imbalance, DC offset, 0–3 tap static multipath—emitting row-aligned $(x_{\text{clean}}, x_{\text{noisy}})$ pairs with all draws recorded. Training windows are drawn by *duration* at random offsets, and empty windows are retained: silence is a label-bearing observation.

5 Dwell-Adaptive Model

A single spectro-temporal encoder (STFT front end; six 3×3 convolution–GroupNorm–SiLU blocks, fully convolutional in time, 32× temporal downsampling; 1.0M parameters alone, 1.57M with the decoder and heads below) feeds three heads (Fig. 3). With episode windows x at dwells drawn from an ordered set $\ell \in \{\ell_1 < \cdots < \ell_K\}$ (we use $\{1, 2.5, 10\}$ ms), spectrogram S_x, random time-frame mask m, and class prototypes computed from support embeddings [3]:

$$\mathcal{L} = \mathcal{L}_{\text{proto}} + \lambda_r \big\| D\big(E(m \odot S_x)\big) - S_{\text{clean}} \big\|_1$$
$$+ \lambda_d(t)\, \mathbb{E}_i \Big[\max_{\ell} \mathcal{L}_{\text{proto}}\big(x_i^{(\ell)}\big) \Big] + \lambda_c\, \mathrm{BCE}\big(g(x), \mathbb{1}[\hat{y} = y]\big),$$

where $\mathcal{L}_{\text{proto}}$ is episodic cross-entropy over squared prototype distances with a learned temperature, $D \circ E$ is the masked denoising path supervised by the clean pair [4, 5], the third term is the worst-case-over-dwell loss (prototypes detached; weight λ_d ramped linearly from zero), and g is a confidence head trained on realized correctness [6, 7]. Optimization uses AdamW with cosine annealing [12].

Dwell policy. At inference the system classifies at the shortest dwell and consults g: if calibrated confidence [9] clears a threshold the answer is emitted; otherwise the capture extends to the next dwell—selective classification [8] where the cost of abstention is observation time, and the analogue of early exiting [10, 11] with physical time replacing depth.

6 Experiments

Setup. Seven classes (`am, bluetooth, cw, dsss, fm, gsm, ofdm`) spanning 34 waveform configurations; 8,704 captures of 20 ms at a common 20 Msps (natively 1.3–122.88 Msps), 96 evaluation rows per configuration (160 further captures per configuration for training); impairments as in Sec. 4. The fixed-count control corpus is that of Sec. 3. Training budget denotes the number of episodic updates (7-way, 5-shot/5-query; dwell drawn uniformly per episode); the base budget is 8,000 episodes and "3×" is 24,000. Optimization: AdamW [12], cosine annealing, peak learning rate $2{\times}10^{-3}$; loss weights λ_r=0.3, λ_d=0.15 (linear ramp over the first tenth of training), λ_c=0.1. Evaluation classifies the capture-initial window of each evaluation row at each dwell, with class prototypes computed from 64 training captures per class under the identical protocol on both corpora.

Duration is the causal factor. Holding the encoder (1.0M parameters, prototype loss only—no reconstruction, worst-case, or confidence terms), optimizer, and budget fixed and swapping only the corpus (Fig. 4): balanced accuracy rises from 0.503 on the shortest fixed-count tier to 0.954 at 1 ms dwell, and from 0.682 to 1.000 at the longest tier vs. 10 ms, with every class

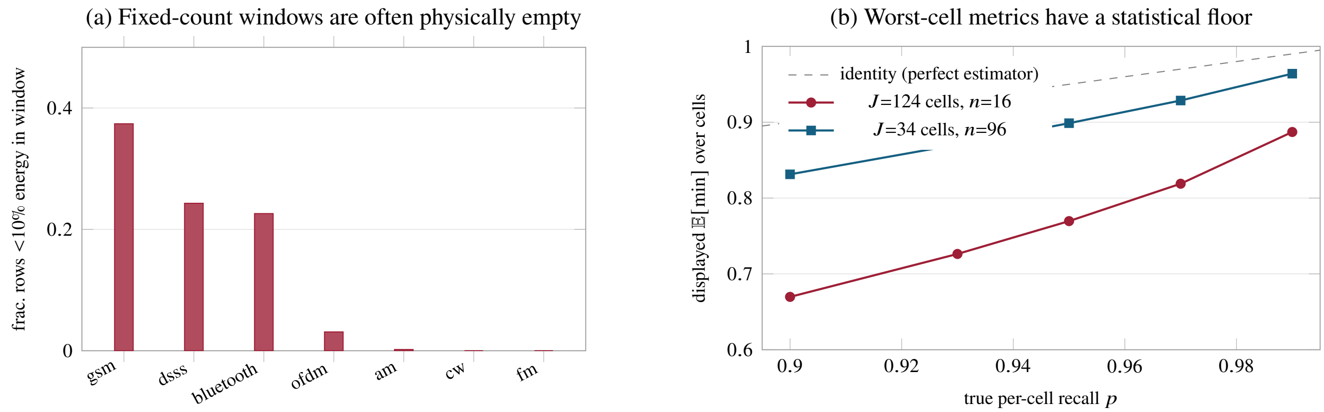

Figure 2: Two failure mechanisms beyond the model. (a) Fraction of fixed-count corpus rows whose scored window contains <10% of capture energy: the hardest classes are frequently *not in the window*. (b) Monte Carlo readout ($2{\times}10^4$ draws) of minimum-over-cells metrics: small-n cells cap what any model can *display*, independent of its true performance.

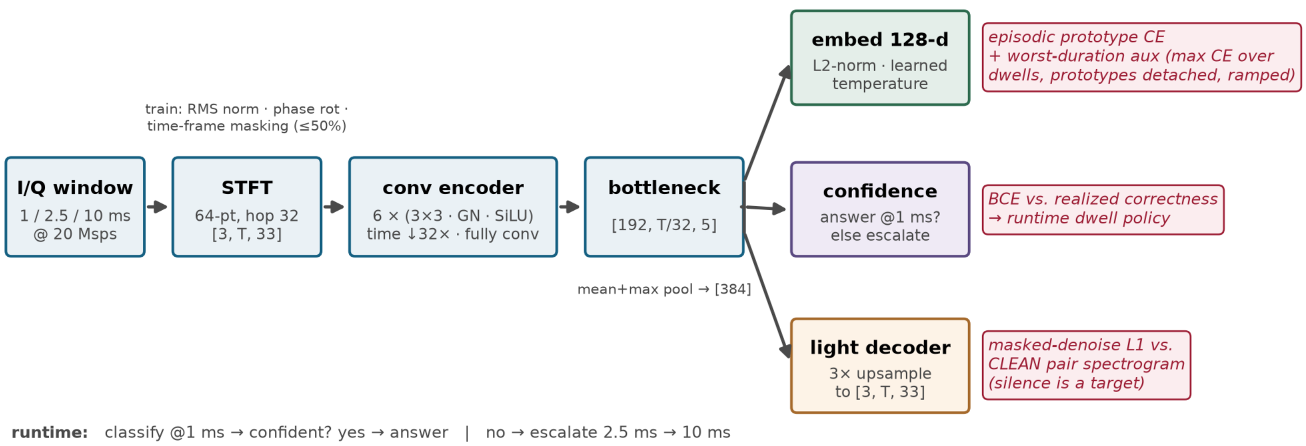

Figure 3: The DACS network. One shared spectro-temporal encoder; three heads. The prototype embedding [3] carries classification and preserves enrollment-based open-set operation; the confidence head [7, 8] drives the answer-early/escalate dwell policy; the light decoder reconstructs the *clean* pair's spectrogram under time-frame masking [4, 5], making silence a learning target and regularizing against the shortcut overfit that duration-starved training exhibits (Sec. 6). The worst-case-over-dwell auxiliary trains the shortest dwell against its hardest cases.

including GSM and Bluetooth at recall 1.000 at 10 ms; an independent seed reproduces 0.958/0.972/1.000. Success criteria fixed before the runs were executed (GSM and BT ≥0.85 at 10 ms; short-dwell balanced ≥ the control level; monotone recall in dwell) all passed on both seeds. Because the control corpus sweeps sample rate, its tiers are matched to dwells by rank (shortest/mid/longest), not by physical duration, which fixed-count data cannot hold constant.

Nothing else closes the gap (Table 1). On the fixed-count control corpus: (i) the same spectro-temporal encoder *loses* to a feature-engineered baseline—hand-designed invariant patch preprocessing (sixteen normalized 64-sample patches with geometry canonicalization) feeding a compact CNN, tuned by a 50-trial hyperparameter search—scoring 0.34 (raw spectrogram input) and 0.45 (adding per-window RMS normalization and phase augmentation) vs. the baseline's 0.49 GSM recall on the shortest tier: architecture is not the lever. (ii) Tripling the training budget *reduces* worst-window balanced accuracy, 0.681→0.657, the signature of shortcut overfitting on ambiguous windows. (iii) A worst-case-over-window auxiliary (max cross-entropy over window tiers) peaks at +0.009 over the 0.681 base-budget baseline (inverted-U dose response, weight ≈0.15) and recovers +0.055 at 3× budget (0.712 vs. 0.657): worst-case pressure helps but cannot supply missing information.

Full system. At an early checkpoint one third through its training schedule, the DACS network reaches balanced accuracy 0.909/0.950/0.990 at 1/2.5/10 ms with minimum-cell recall 0.448/0.760/0.948; surviving errors concentrate in single configurations at the shortest dwell—precisely the cases escalation targets. The raw confidence head predicts the answer-at-1 ms decision at 0.884 accuracy against a 0.909 marginal correctness rate at this checkpoint; the head is uncalibrated here, and we accordingly claim the escalation *mechanism*, deferring its operating curve to calibration on held-out data (Sec. 7). Training cost is linear in dwell: 1.4 ms (1 ms window) to 31 ms (20 ms window) per training step per window on one Apple M-series GPU, so duration completeness is computationally cheap.

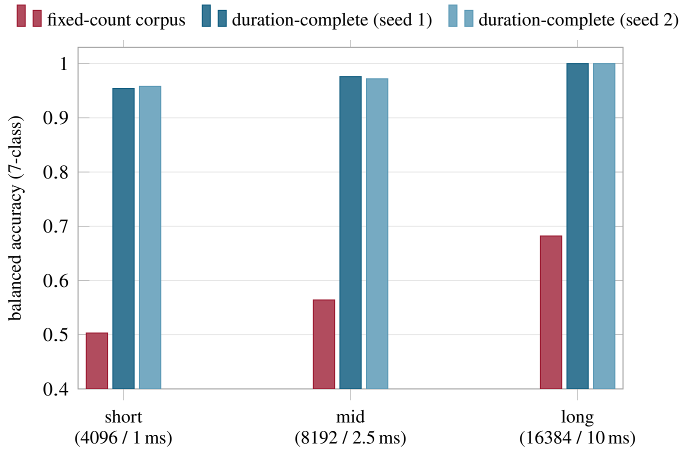

Figure 4: Controlled attribution. The identical 1.0M-parameter encoder, recipe, and budget on the fixed-count control corpus (window tiers in samples) vs. duration-complete data (dwells in milliseconds); tiers are matched by rank, not physical duration, which fixed-count data cannot hold constant. At 10 ms every class reaches recall 1.000 on both seeds.

Table 1: Controls on the fixed-count corpus: no model-side intervention approaches the effect of duration-complete data.

Intervention (control corpus)	Metric	Effect
Feature-eng. baseline, 50-trial search	worst-win. bal.	0.743
its GSM recall, shortest tier	GSM recall	0.49
Our encoder, raw / normalized input	GSM recall	0.34 / 0.45
3× training budget	worst-win. bal.	0.681 → 0.657
Worst-case loss, wt. .05/.15/.40	Δ w.-win.	+.003/+.009/−.008
Worst-case loss, 3× budget	Δ w.-win.	+.055
Duration-complete data, fixed encoder	shortest-tier bal.	0.503 → 0.954

7 Limitations

All data are synthetic; over-the-air validation remains future work [2]. The hop composition reproduces occupancy and timing statistics, not the specification's selection kernel, and repeats verified payload bits. Captures contain a single emitter; co-channel interference and Doppler are absent. Confidence calibration [9] on held-out data is required before escalation thresholds are meaningful, and the full-system numbers above are an early snapshot at one third of the training schedule.

8 Conclusion

For burst-structured RF, the binding constraint on classification is what the training window can physically contain—and secondarily, what the evaluation can statistically display. DACS makes capture duration a controlled property of synthesis, teaches silence as evidence, trains the shortest dwell against its hardest cases, and is designed to spend observation time only where confidence demands it. The controlled result—0.503 to 0.954 at 1 ms with architecture fixed—suggests a general prescription: before scaling models on a plateaued worst-case metric, audit the observation window and the metric itself.

References

[1] T. J. O'Shea, J. Corgan, and T. C. Clancy, "Convolutional radio modulation recognition networks," in *Proc. EANN*, 2016.

[2] T. J. O'Shea, T. Roy, and T. C. Clancy, "Over-the-air deep learning based radio signal classification," *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 1, 2018.

[3] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *NeurIPS*, 2017.

[4] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *CVPR*, 2022.

[5] P. Vincent, H. Larochelle, I. Lajoie, Y. Bengio, and P.-A. Manzagol, "Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion," *J. Mach. Learn. Res.*, vol. 11, 2010.

[6] T. DeVries and G. W. Taylor, "Learning confidence for out-of-distribution detection in neural networks," arXiv:1802.04865, 2018.

[7] D. Hendrycks and K. Gimpel, "A baseline for detecting misclassified and out-of-distribution examples in neural networks," in *ICLR*, 2017.

[8] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *NeurIPS*, 2017.

[9] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *ICML*, 2017.

[10] S. Teerapittayanon, B. McDanel, and H. T. Kung, "BranchyNet: Fast inference via early exiting from deep neural networks," in *ICPR*, 2016.

[11] G. Huang, D. Chen, T. Li, F. Wu, L. van der Maaten, and K. Q. Weinberger, "Multi-scale dense networks for resource efficient image classification," in *ICLR*, 2018.

[12] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *ICLR*, 2019.

[13] Bluetooth SIG, *Bluetooth Core Specification*, Version 6.3, 2026.

[14] 3GPP, "Multiplexing and multiple access on the radio path," TS 45.002, Rel. 19, 2026.